

No beast on Earth is tougher than the tiny *tardigrade* (缓步类动物). It can 26 being frozen at -272° Celsius, being exposed to the vacuum of outer space and even being 27 with 500 times the dose of X-rays that would kill a human. In other words, the creature can endure conditions that don't even exist on Earth. And researchers are looking to the microscopic animals to learn how to prepare humans and crops to handle the 28 of space travel.

The tardigrade's indestructibility stems from its 29 to its environment—which may seem surprising, since it lives in 30 comfortable places, like the cool, wet patches of *moss* (青苔) that dot a garden wall.

But it turns out that a tardigrade's damp, mossy home can dry out many times each year. Drying is pretty 31 for most living things. It does damage to cells in some of the same ways that freezing, vacuum and radiation do. Tardigrades, however, have 32 special strategies for dealing with these kinds of damage.

As a tardigrade dries out, its cells produce several strange proteins that are unlike anything found in other animals. In water, the proteins are shapeless. But as water disappears, the proteins self-assemble into long fibers that fill the cell's 33. The fibers support the cell's *membranes* (细胞膜) and proteins, preventing them from breaking or 34.

Emulating tardigrades could one day help humans colonize outer space. Food crops could be engineered to produce tardigrade proteins, allowing these organisms to grow more efficiently on spacecraft where levels of radiation are elevated compared with on Earth.

So if humans ever succeed in reaching the stars, they may accomplish this 35, in part, by standing on the shoulders of the tiny eight-legged endurance specialists in your backyard.

A) adaptations

B) blasted

C) catastrophic

D) evolved

E) feat

F) interior

G) probing

H) recurrence

I) rigors

J) seemingly

K) survive

L) tempt

M) thrill

N) unanimously

O) unfolding